

IEEE MTI STUDENT BRANCH

HOUSE PRICE PREDICTION

Graduation Project: California Housing Market Intelligence

2026 graduation patch

Development Team

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Introduction

The Market Challenge

The California real estate market is defined by high volatility and complex socio-economic correlations.

Traditional valuation methods often fail to scale with rapid market shifts. Our objective is to engineer a **scientific, automated valuation system** using Big Data to provide objective market value estimations.



Project Objectives



Big Data Mining

Analyzing 20,640 records to find hidden district-level price correlations.



Model Benchmarking

Developing optimized regression models to minimize estimation error.



Interactive UI

Deploying a web-ready dashboard for real-time housing valuation.

Dataset Anatomy

Parameter	Type	Impact on Valuation
Median Income	Numerical	Strongest driver of regional housing value.
Ocean Proximity	Categorical	Geographic premium based on coastal distance.
Housing Median Age	Numerical	Determines property depreciation influence.
Longitude / Latitude	Coordinate	Used for district-level spatial clustering.
Total Rooms	Numerical	Reflects district residential capacity.

DATA REFINEMENT

From Raw Data to Algorithmic Power

Preprocessing Workflow



Cleaning

Imputed 207 missing entries in 'total_bedrooms' using median strategy to maintain data integrity.



Standardization

Applied StandardScaler to ensure uniform feature magnitude across all predictors.



Encoding

Converted categorical 'ocean_proximity' into binary vectors via One-Hot Encoding.

Feature Engineering

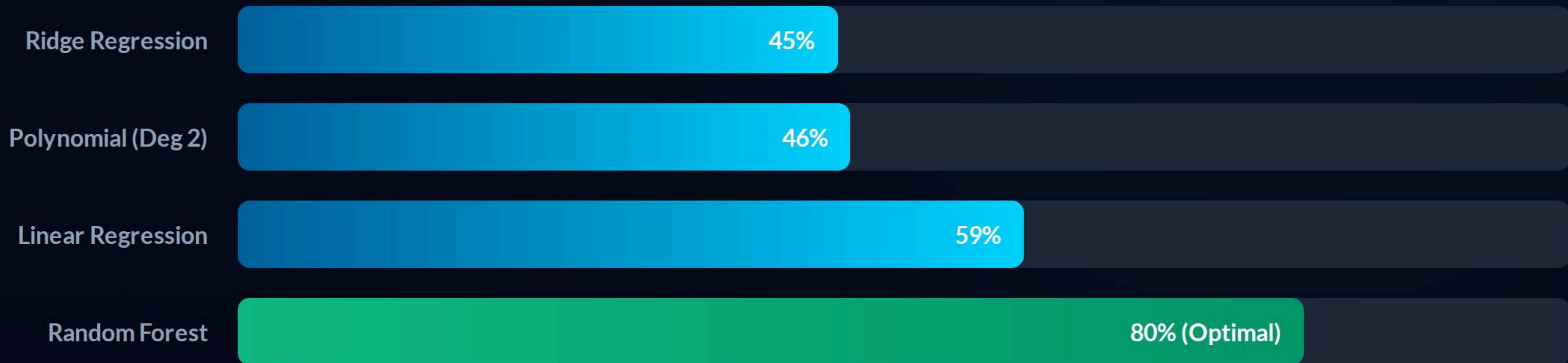


Creating New Insights

We engineered synthetic features to capture deep district-level density:

- ✓ Rooms per Household
- ✓ Population per Household
- ✓ Bedrooms per Room Ratio

Performance Benchmarks



Random Forest outperformed all parametric models by capturing non-linear spatial complexities.

Winning Logic

Random Forest Regressor

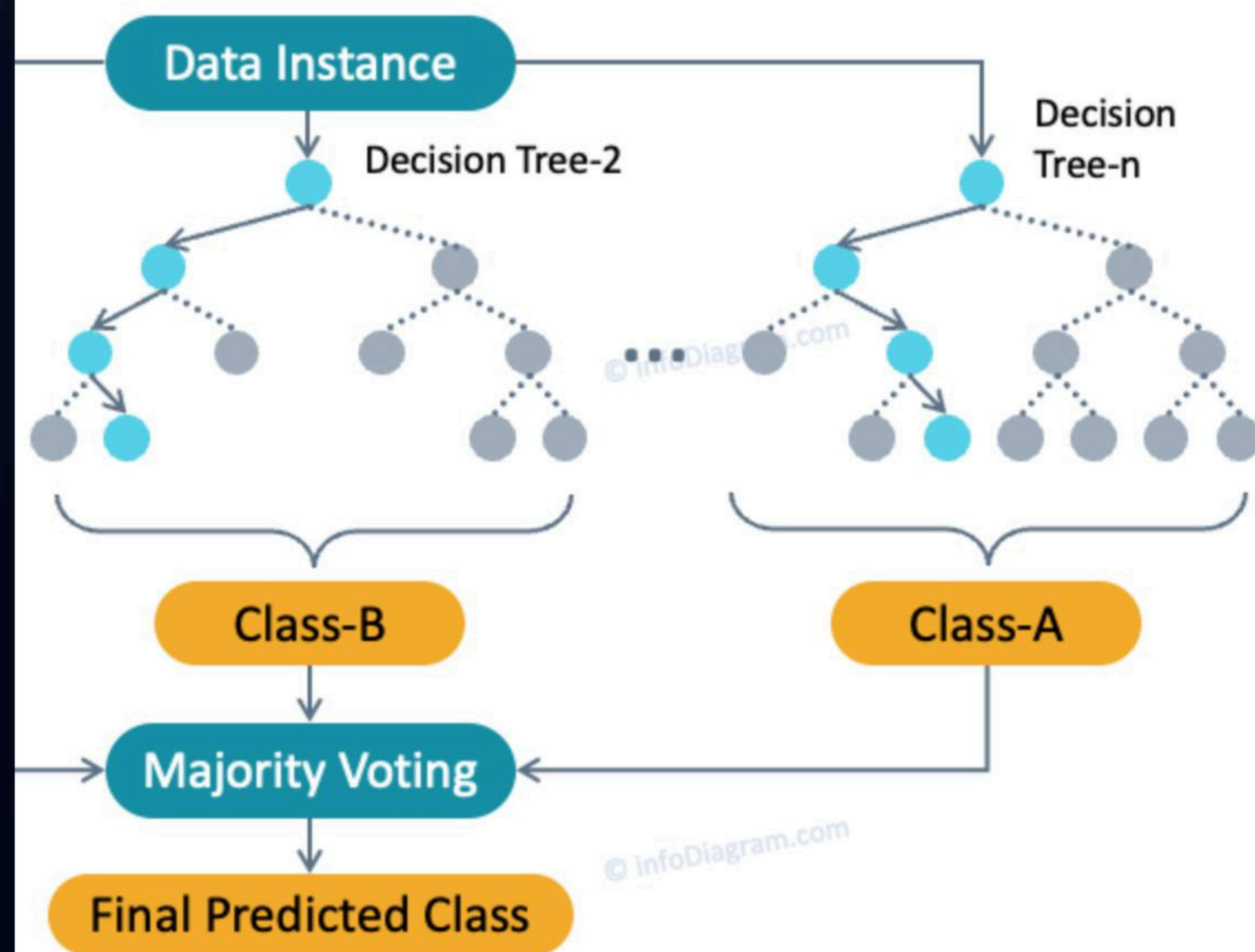
By aggregating the predictions of hundreds of decision trees, the model effectively reduces variance and handles outliers in the 20,640 records.

This ensemble approach proved essential for the non-linear nature of California's diverse real estate market.

Best Illustration - Supervised Learning

Flowchart Diagram, Decision Tree Set, Predicted Class, Voting

Machine Learning: Random Forest



Model Reliability

0.80

R^2 ACCURACY SCORE

Predictive Success

The final model explains **80% of the price variance** in the California dataset.

"A high-precision result for complex regional real estate valuation."

Live Deployment

Production Implementation

We deployed the winning model using **Gradio**, creating a live dashboard for stakeholders to:

- ✓ Input demographics via interactive sliders.
- ✓ Select geographic proximity parameters.
- ✓ Receive instantaneous price estimations.





THANK YOU

Questions or Discussion?

IEEE MTI Student Branch | Graduation Project 2024

Image Sources



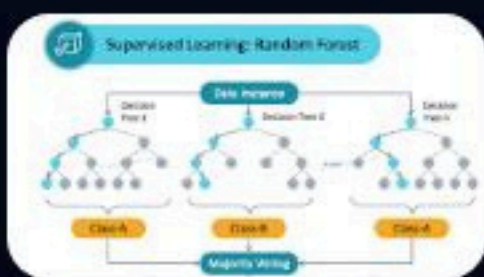
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